

PROJECT REPORT

ON

CRIME REPORTING SYSTEM

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**UTTARANCHAL
UNIVERSITY
DEHRADUN**

**NAAC
GRADE A+**

Under the supervision of:

Dr. Manoj Aswal

(Assistant Professor)

USCS

Submitted By:

Simran

Enrollment No:

UU2509000320

Acknowledgement

The most awaited moment of any endeavor is its successful completion, but nothing can be done successfully if done alone. Success is the outcome of the contribution and consistent help of various persons, and we thank those who helped us complete this project. Primarily, I would like to thank **Prof. (Dr.) Sonal Sharma**, Director of USCS, for providing a healthy and encouraging environment in which to study.

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Certificate of completion

This is to certify that Simran student of BCA 2nd Semester of Uttarakhand School of Computing Sciences, Uttarakhand University, Dehradun, has completed the Project Report on Crime Reporting System in C++ Language for the Batch (2025-2028).

Dept. of Computing Sciences USCS,

Under the Guidance of

Dr. Manoj Aswal

Uttarakhand University

Declaration

I hereby declare that the project report entitled Crime Reporting system by Simran to Uttarakhand School of Computing Sciences, Uttarakhand University. The project was done under the guidance of Dr. Manoj Aswal. I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this university or any other university.

Name of Student:

Simran

(Signature)

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Introduction

A **Crime Reporting System** is a software application designed to help law enforcement agencies or organizations **record, manage, and track crime-related information efficiently**. Traditionally, crime records were maintained manually in registers and files, which made searching, updating, and maintaining records difficult and time-consuming. By using a computerized system developed with C++, crime data can be stored, processed, and retrieved quickly and accurately.

A Crime Reporting System developed in C++ **programming language** uses concepts such as **structures, classes, file handling, and object-oriented programming** to manage crime records. The system allows users or administrators to **add new crime reports, view existing records, update information, delete cases, and search for specific reports** based on details like case ID, criminal name, location, or date.

The main goal of this system is to **improve efficiency, maintain proper records, reduce paperwork, and ensure better data management**. C++ is suitable for such projects because it is **fast, reliable, and supports object-oriented programming**, which helps organize large programs into manageable modules.

Objective

1. To develop an online **Crime Reporting System** that allows citizens to report crimes easily and enables authorities to manage and respond to complaints efficiently.
2. **To provide an easy platform for citizens** to report crimes online without visiting a police station.
3. **To maintain a centralized database** of crime reports for better record management.
4. **To allow users to upload details and evidence** such as images, location, and description of the incident.
5. **To help police authorities track and manage complaints** through an organized dashboard.

System Analysis

1. Identification of Need:

The traditional method of reporting crimes requires citizens to visit police stations and file complaints manually. This process is time-consuming and inconvenient. Therefore, a computerized Crime Reporting System is needed to allow citizens to report crimes online and maintain records efficiently.

2. Preliminary Investigation:

The investigation of the existing system shows that crime records are maintained manually, which leads to excessive paperwork, difficulty in searching records, and chances of data loss. A digital system can solve these problems by storing and managing data electronically.

Feasibility Study

○Technical Feasibility: The system can be developed using common technologies (web/mobile apps, database, cloud servers). Required tools and skills are widely available.

○Economic Feasibility: initial Development and maintenance costs are moderate, but it reduces paperwork and improves efficiency, saving costs long term.

○Operational Feasibility: Easy for citizens use and police can adopt it with basic training.

Improve communication and response time.

○Legal Feasibility: Must follow data privacy and cybersecurity laws. With proper security measures, compliance is achievable.

SDLC (Software Development Life Cycle)

1. **Planning** – Define goals, scope, and resources for the system.
2. **Requirement Analysis** – Collect user needs like reporting crimes, tracking complaints, and uploading evidence.
3. **Design** – Plan system structure, database, and user interface.
4. **Development** – Build the system (coding frontend, backend, and database).
5. **Testing** – Check for bugs, errors, and security issues.
6. **Deployment** – Launch the system for public and police use.
7. **Maintenance** – Update, fix issues, and improve features over time.

System Requirements

Functional Requirements:

Here's a **version of the functional requirements** for a Crime Reporting System:

1. **User Registration/Login** – Citizens and police can securely register and log in.
2. **Crime Reporting** – Citizens can file crime complaints with details and evidence.
3. **FIR Generation** – System automatically creates a First Information Report for each complaint.
4. **Case Management** – Police can view, update, and assign cases.
5. **Notifications** – Users get alerts about report submission and case status.
6. **Tracking & Search** – Citizens track their complaints; police search cases by type, location, or status.
7. **Reports & Analytics** – Generate crime statistics and trends.
8. **Security** – Role-based access control and data protection.

Non-Functional Requirements (How the system should work):

- **Performance** – Fast response and handles multiple users.
- **Reliability** – Available 24/7 with minimal downtime.
- **Security** – Data protection, authentication, and role-based access.
- **Usability** – Easy-to-use interface for citizens and police.
- **Maintainability** – Easy to update and fix issues.
- **Scalability** – Can handle more users and data as needed.

Hardware and Software

Hardware Requirements

- **Processor:** Intel Core i5 or higher
- **RAM:** 4 GB or more
- **Hard Disk:** 120–500 GB storage
- **Monitor:** 15-inch or larger
- **Input Devices:** Keyboard and Mouse
- **Network:** Internet connection for online reporting
- **Computer:** Desktop or Laptop

Software Requirements

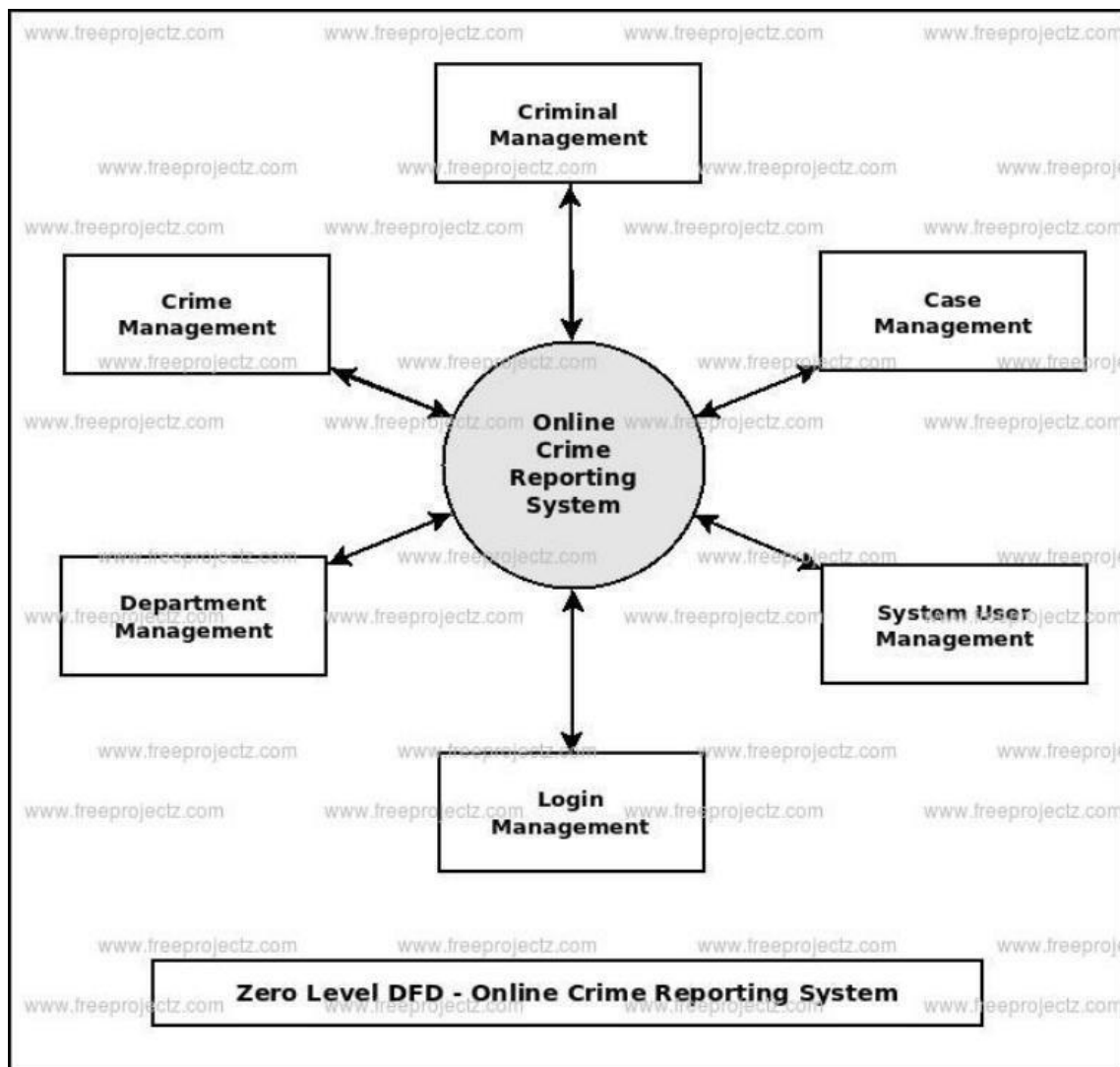
- **Operating System:** Windows 11
- **Programming Language:** C++, C
- **Development Tools/IDE:** Visual Studio, Dev C++
- **Web Browser:** Chrome, Firefox, or Edge

Data Flow Diagram (DFD)

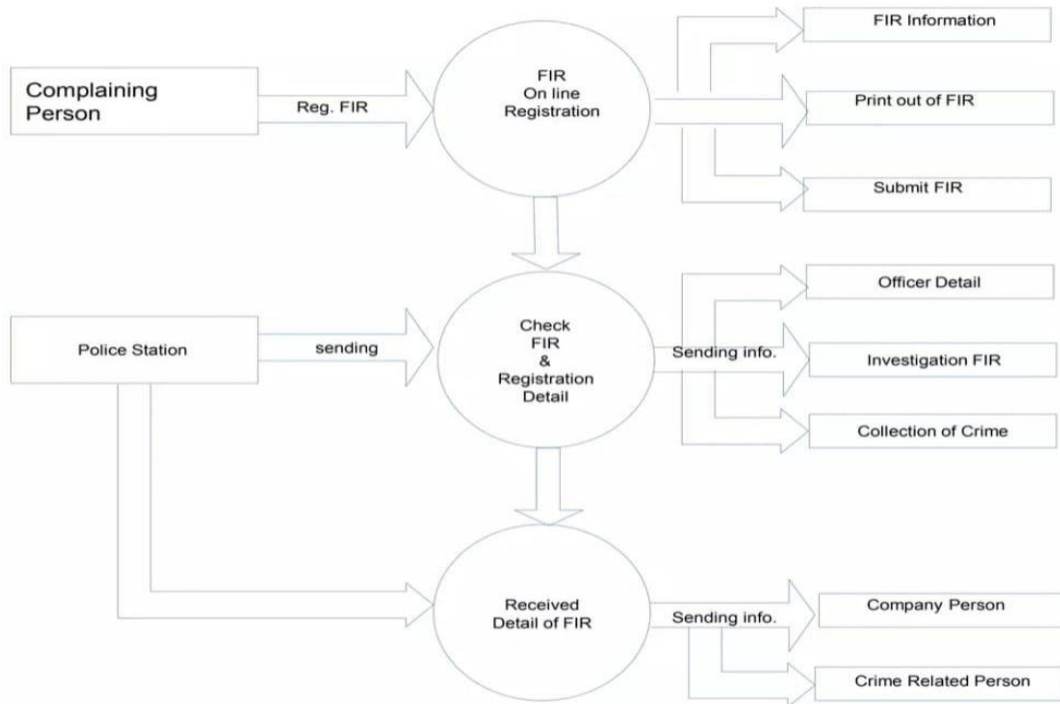
- **Level 0 DFD**

This shows the whole system as a single process.

[Citizen] ----->
 (Crime Reporting System)
[Police/Admin] <-----



- **Level 1 DFD**



1.

External Entities

- Citizen: Reports crimes, checks status
- Police/Admin: Investigates and updates cases

2. Processes

- Submit Crime Report: Citizen enters details like location, type of crime
- Verify Report: System checks validity (duplicate, incomplete data)
- Assign Case: Police assigns officer for investigation
- Update Status: Case progress is updated

3. Data Store

- Crime Report Database: Stores all reported cases and evidence

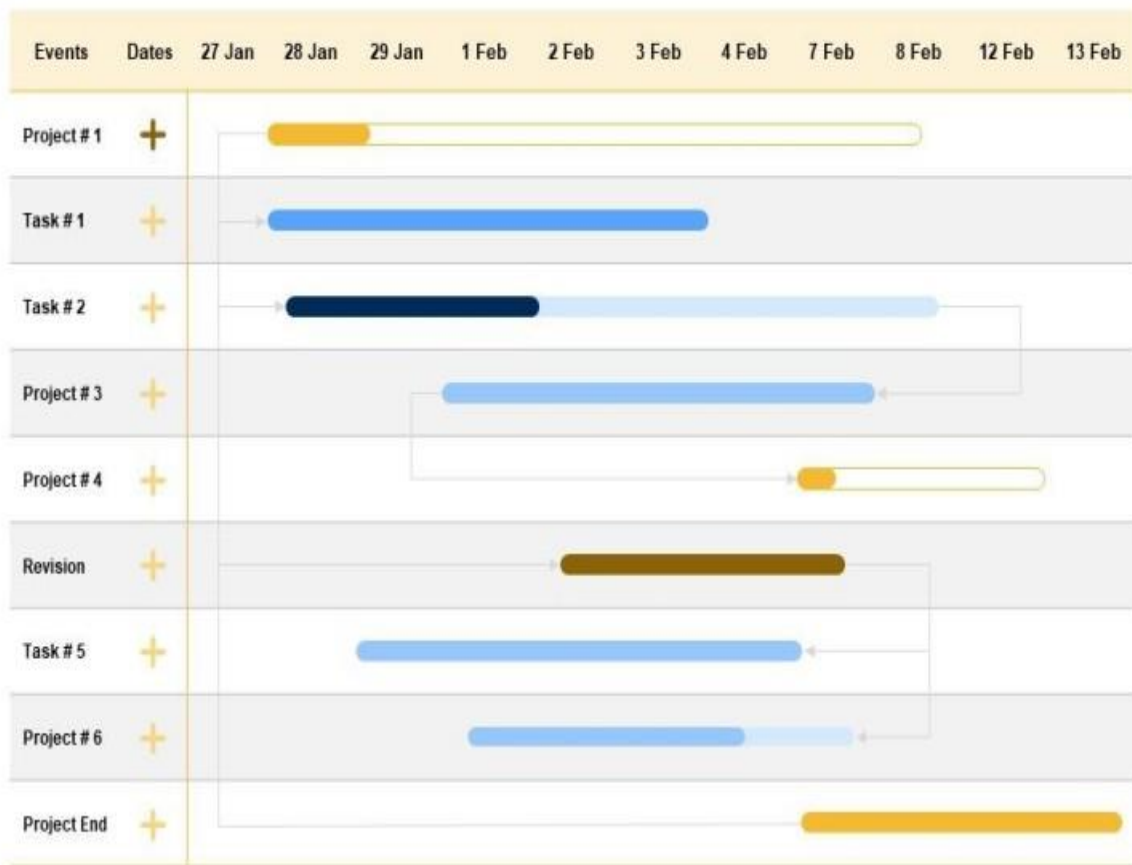
4. Data Flow

- Crime details → system
- Verified report → police

- Status updates → citizen

Gantt Chart Diagram

Gantt Chart for Security Project Proposal

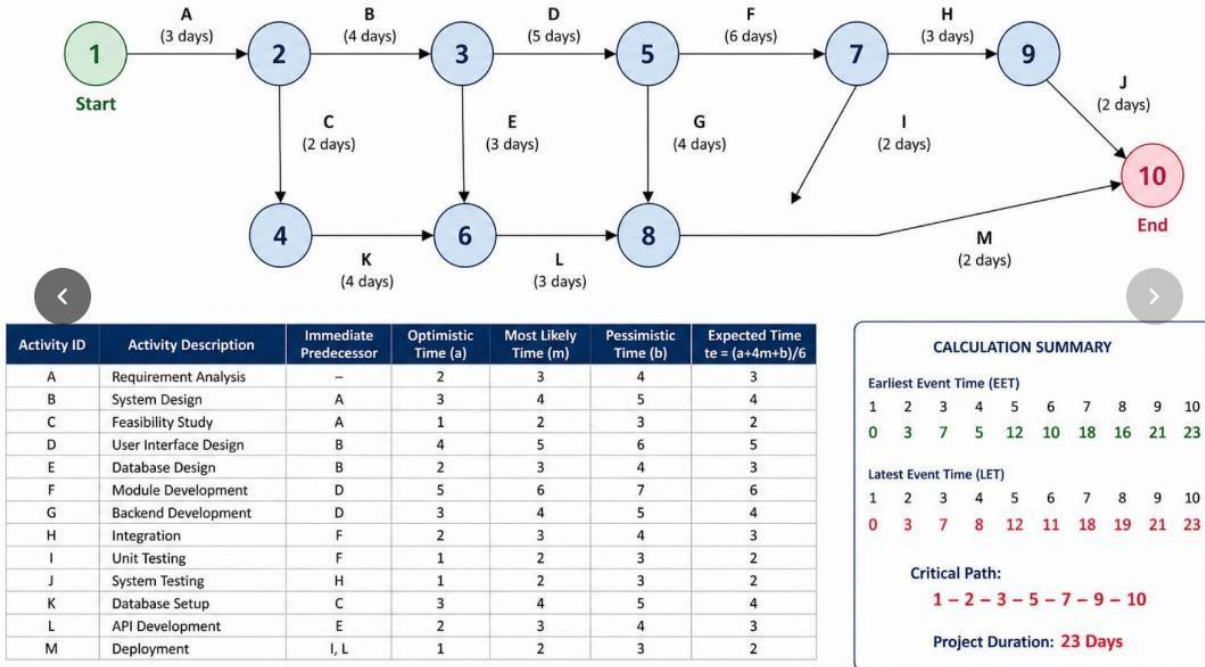


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Pert chart Diagram

PERT CHART – CRIME REPORTING SYSTEM



Advantages

- ☐ **Quick reporting** – Crimes can be reported instantly without visiting a police station.
- ☐ **Faster response time** – Authorities can act more quickly on reported incidents.
- ☐ **24/7 availability** – Systems are accessible anytime, increasing convenience.
- ☐ **Easy accessibility** – Can be used by anyone with a phone or internet connection.
- ☐ ☐ **Better data management** – Stores records in an organized digital format.

Limitations

- **Internet Dependency** – Citizens need internet access to report crimes online.
- **Technical Skills Required** – Some users may find the system difficult if they aren't techs.
- **Data Privacy Risks** – Sensitive information could be at risk if security is breached.
- **Limited Offline Access** – Cannot report or track cases without network connectivity.

Algorithm

1. **Start the system** and open login/registration page.
2. **User Registration/Login**
 - New user registers with name, email, and password.
 - Existing user logs in using credentials.
 - System verifies details.
3. **User Dashboard**
 - After login, user sees options:
 - Report Crime
 - View Complaint Status
 - Update Profile
 - Logout
4. **Report Crime**
 - User enters crime details (type, description, location, date/time).
 - System validates input.
 - System generates a unique Complaint ID.
 - Complaint is stored in the database with status “Pending”.
5. **Admin Processing**
 - Admin logs in and views complaints.
 - Admin verifies complaint details.
 - Admin updates status (Pending → Under Investigation → Resolved/Rejected).
6. **View Status**
 - User enters Complaint ID.
 - System displays current status and details.

7. **Logout** ○ User/Admin logs out.

○ Session ends.

8. End.

Future Scopes and Enhancement of Project

- ❑ **Mobile & Online Access** – Citizens can report and track crimes via apps or web.
- ❑ **Real-Time Notifications** – Alerts via SMS, email, or app for updates.
- ❑ **AI & Analytics** – Predict crime patterns and identify high-risk areas.
- ❑ **Integration with Emergency Services** – Quick response from police, ambulance, or fire.
- ❑ **Enhanced Features** – Geo-tagging, digital evidence uploads, chatbots, and role-based dashboards.

Role of Technology In Crime Reporting System

The role of technology in a Crime Reporting System is to make reporting, tracking, and managing crimes faster, easier, and more secure using digital tools.

- **Online Reporting** – Citizens can file complaints via web or mobile.
- **Faster Case Management** – Police can track and update cases in real-time.
- **Data Storage & Retrieval** – Securely store and access crime records.
- **Notifications** – Automatic alerts about case status.

Methodology

The methodology for a Crime Reporting System involves several structured steps to ensure efficient development and implementation. First, requirement analysis is conducted to understand user needs, including citizens and law enforcement agencies. Then, the system is designed with appropriate architecture, database structure, and user-friendly interfaces. During development, technologies like web or mobile platforms are used to build modules such as user registration, crime reporting, and case tracking. The system is then tested through various methods to ensure accuracy, security, and performance. After successful testing, it is deployed on a server for realworld use. Finally, regular maintenance is carried out to fix issues, update features, and ensure data security and reliability.

Recommendations

A system that allows crimes to be reported, tracked, and managed electronically.

- ☐ Ensure a **user-friendly interface** for easy reporting.
- ☐ Implement **strong security** with encryption and access control.
- ☐ Provide **mobile access** for reporting and tracking.
- ☐ Enable **real-time notifications** for updates.
- ☐ Conduct **regular maintenance and updates**.

Conclusion

The **Crime Reporting System** represents a significant advancement in the way citizens interact with law enforcement. By leveraging modern technology, it allows users to **report crimes quickly and securely from anywhere**, eliminating the need to physically visit a police station. The system improves **efficiency and transparency** by enabling police officers to **manage, track, and update cases in real-time**, ensuring timely action and better coordination.

With features like **FIR generation, notifications, case tracking, and digital evidence management**, the system not only reduces manual paperwork but also helps in **organizing crime data systematically**. This data can be further used for **analytics and crime prediction**, assisting authorities in identifying high-risk areas and taking preventive measures.

Moreover, the system emphasizes **security and privacy**, protecting sensitive information through **encryption, authentication, and role-based access**, while also providing a **user-friendly interface** for citizens of varying technical abilities. The integration with **emergency services** and potential for **mobile access** ensures that the system is not only reactive but also proactive in safeguarding communities.

Source Code

```
#include <iostream>

#include <fstream>

#include <string> using
namespace std;

class CrimeReport { public:
    int id;    string title;
    string description;
    string location;

    void input() {        cout <<
"Enter Report ID: ";        cin >>
id;        cin.ignore();

        cout << "Enter Crime Title: ";
getline(cin, title);

        cout << "Enter Description: ";
getline(cin, description);

        cout << "Enter Location: ";
getline(cin, location);
    }

    void display() {        cout << "\nReport ID: "
```

```

<< id;    cout << "\nTitle: " << title;    cout <<
"\nDescription: " << description;    cout
<< "\nLocation: " << location << endl;    cout
<< "-----\n";
    }
};

/ Add report void addReport() {
CrimeReport c;    ofstream
file("crime.txt", ios::app);

    c.input();

    file << c.id << endl;    file
<< c.title << endl;    file <<
c.description << endl;    file <<
c.location << endl;

    file.close();

    cout << "\nReport Added Successfully!\n";
}

// View all reports void
viewReports() {
ifstream
file("crime.txt");

    CrimeReport c;

```

```

        while (file >> c.id) {
file.ignore();    getline(file,
c.title);    getline(file,
c.description);    getline(file,
c.location);

        c.display();
    }

    file.close();
}

// Search report by ID void
searchReport() {    ifstream
file("crime.txt");
CrimeReport c;    int
searchId;    bool found =
false;

    cout << "Enter Report ID to search: ";
cin >> searchId;

    while (file >> c.id) {
file.ignore();    getline(file,
c.title);    getline(file,
c.description);    getline(file,
c.location);

```

```

        if (c.id == searchId) {
            c.display();
found = true;
break;
        }
    }

    if (!found) {        cout <<
"Report not found!\n";
    }

    file.close();
}

int main() {    int
choice;

    do {        cout << "\n===== Crime Reporting System
=====\\n";        cout << "1. Add Report\\n";        cout <<
"2. View Reports\\n";        cout << "3. Search Report\\n";
cout << "4. Exit\\n";        cout << "Enter choice: ";        cin
>> choice;

        switch (choice) {            case 1:
addReport();            break;
case 2:            viewReports();
break;            case 3:

```

```
searchReport();      break;
case 4:      cout << "Exiting...\n";
break;      default:      cout <<
"Invalid choice!\n";
    }

} while (choice != 4);

return 0;
}
```

Output

```
===== Crime Reporting System =====
1. Add Report
2. View Reports
3. Search Report
4. Exit
Enter choice: 1
Enter Report ID: 101
Enter Crime Title: Murder
Enter Description: Person found dead at home with stab wounds, suspected murder
Enter Location: Rohini, Delhi

Report Added Successfully!
```

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